MODULE 2

Descriptive Statistics & Data Analysis

DBS-PC-BR-601-T · Business Research and Statistics · MBA Semester I

Excel-based · Anderson, Sweeney et al. Ch. 2–3 | Levine, Szabat & Stephan Ch. 2–3

What you will be able to do

Learning outcomes for this module

Summarise the centre

Compute and choose between the mean, median and mode for business data.

Measure the spread

Use range, IQR, variance, standard deviation and the coefficient of variation.

Describe the shape

Read skewness and kurtosis, and detect outliers with z-scores.

Quantify association

Compute and interpret Pearson and Spearman correlation.

Choose the right chart

Match bar, pie, Pareto, histogram, box plot and scatter to the data type.

Do it all in Excel

Descriptive Statistics tool, PivotTables and the core statistical functions.

Module roadmap

Five topics, each with an Excel workout

1	2	3	4	5
Central tendency	**Dispersion**	**Shape**	**Association**	**Visualisation**
Mean, median, mode — and when each one lies	Range, IQR, variance, SD, CV, box plots	Skewness, kurtosis, outliers, z-scores	Covariance, Pearson r, Spearman rank	Choosing and building the right chart

Everything in this module describes the data you actually hold. Module 3 uses these same numbers to reach beyond the sample.

1

Measures of Central Tendency

One number to stand for the whole distribution

Mean, median and mode

Three answers to ‘what is typical?’

MEAN

Add every value, divide by the count.

Uses all the data. Pulled hard by extreme values.

`=AVERAGE(range)`

MEDIAN

The middle value once the data are sorted.

Ignores how extreme the extremes are. Resistant.

`=MEDIAN(range)`

MODE

The value that occurs most often.

The only one that works for categorical data.

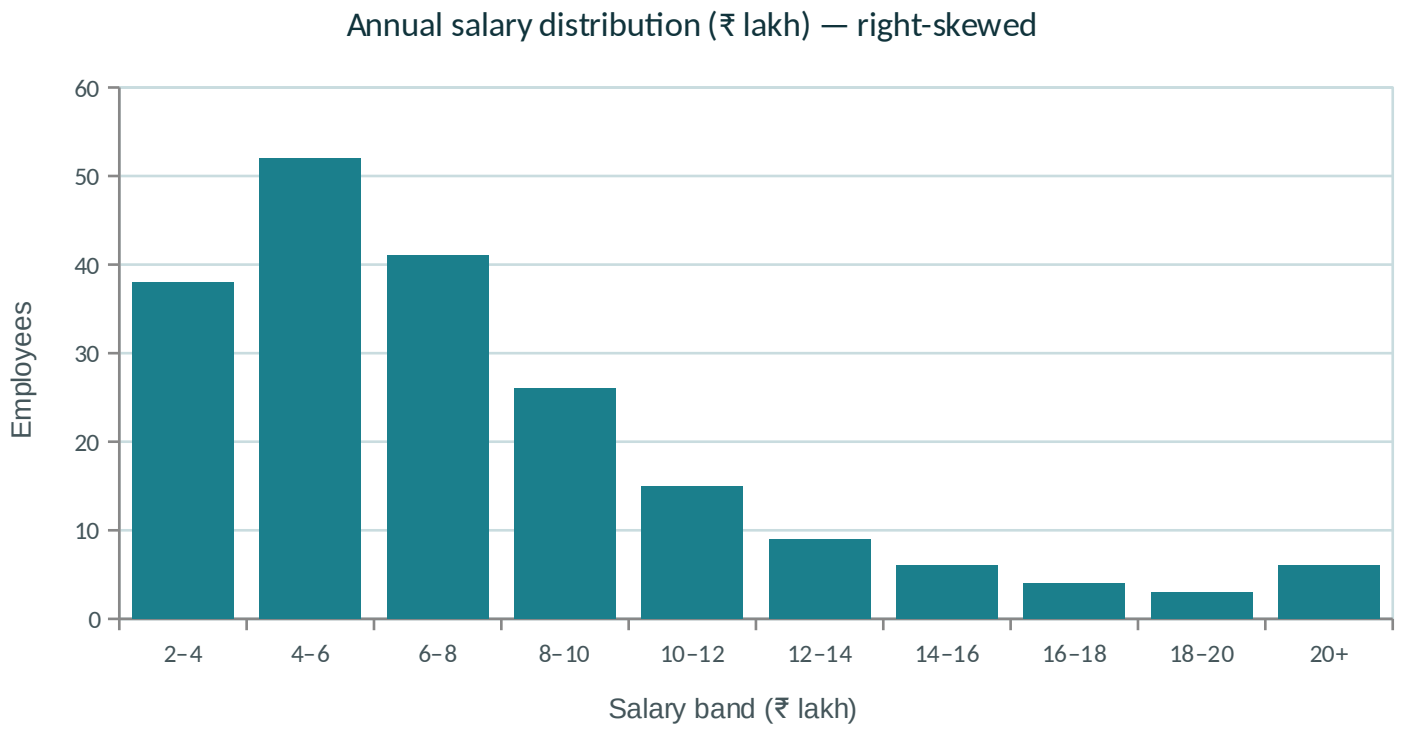
`=MODE.SNGL(range)`

Salaries of 5 employees (₹ lakh): 4.2 4.8 5.1 5.6 28.0

Mean = ₹9.54 lakh. Median = ₹5.10 lakh. Four of the five employees earn less than the mean — the founder’s salary has dragged it away from anything typical.

When the mean lies

In a skewed distribution the mean and the median part company



Median ≈ ₹5.6 lakh **Mean ≈ ₹7.4 lakh** — the long right tail pulls the mean away from the typical employee.

Which one do you report?

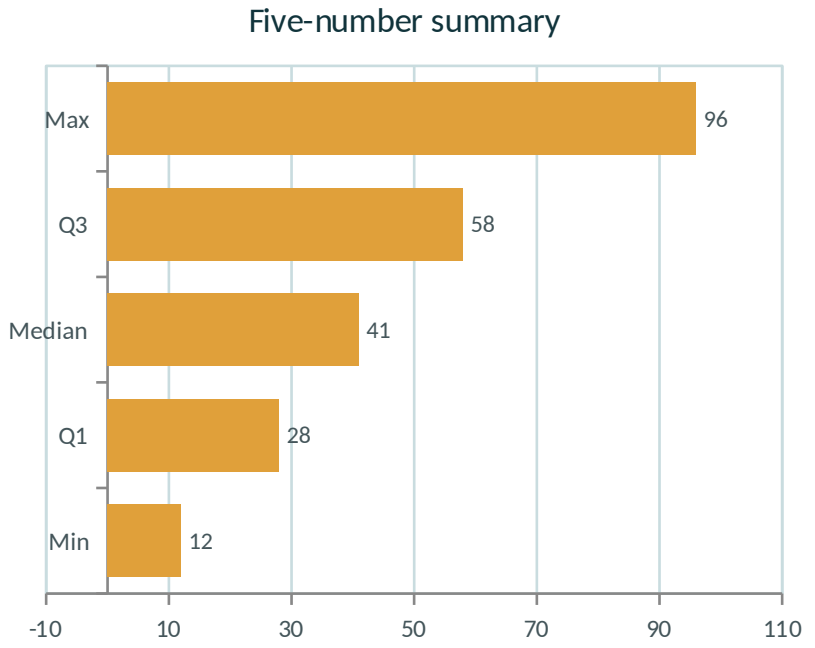
- Roughly symmetric data → the mean. It uses every observation.
- Skewed data → the median. Incomes, house prices, claim sizes and waiting times are almost always skewed.
- Categorical data → the mode. It is the only one defined.
- In doubt, report both and let the gap between them tell the story.

Illustrative distribution

Percentiles and quartiles

Cutting the distribution into ordered pieces

Q1	**25th percentile**	25% of observations fall below this value	`=QUARTILE.EXC(range, 1)`
Q2	**50th percentile**	The median. Half the data lie on each side	`=MEDIAN(range)`
Q3	**75th percentile**	75% of observations fall below this value	`=QUARTILE.EXC(range, 3)`
IQR	**Q3 − Q1**	The range of the middle half — the resistant measure of spread	`=Q3−Q1`



These five numbers are exactly what a box plot draws.

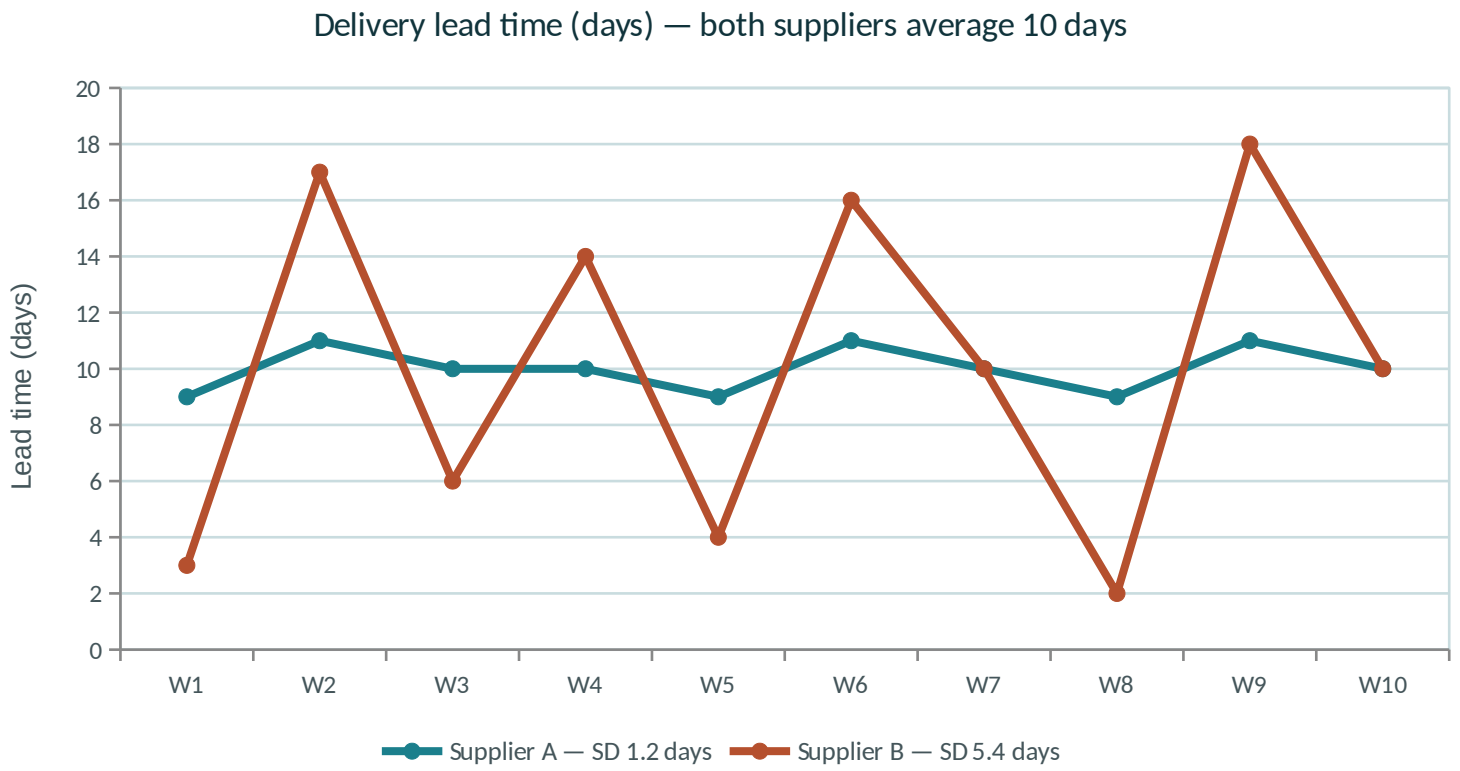
2

Measures of Dispersion

Two products with the same average can be entirely different businesses

Why the average is never enough

Same mean, different risk



Illustrative data

Identical mean

Both suppliers average 10 days. On the mean alone you would be indifferent between them.

Opposite risk

Supplier B forces you to hold far more safety stock. Variability, not the average, drives the inventory decision — which is why dispersion is measured before anything else.

The four measures of spread

From crude to complete

1

Range

Max − Min

Instant, but decided entirely by the two most extreme observations.

```
=MAX()−MIN()
```

2

Interquartile range

Q3 − Q1

The spread of the middle 50%. Resistant to outliers.

```
=QUARTILE.EXC(r,3)−QUARTILE.EXC(r,1)
```

3

Variance

Mean squared deviation from the mean

Uses every value. Its units are squared, so it is hard to read directly.

```
=VAR.S(range)
```

4

Standard deviation

The square root of the variance

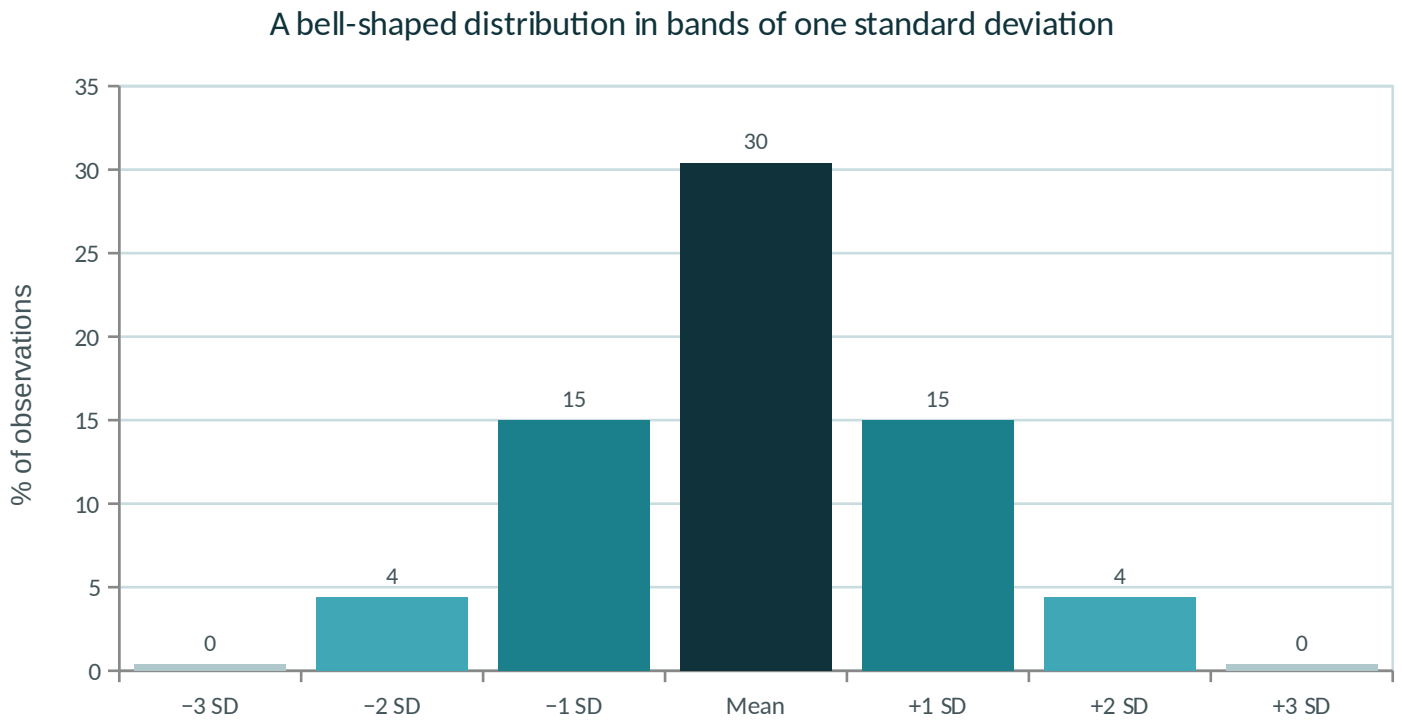
Back in the original units — the one you report.

```
=STDEV.S(range)
```

Coefficient of variation = standard deviation ÷ mean, as a percentage. Use it whenever you compare the variability of two things measured in different units or on very different scales.

The empirical rule

For roughly bell-shaped data, the standard deviation maps the whole distribution



± 1 SD **≈68%**

About two-thirds of all observations

± 2 SD **≈95%**

The band you quote to management

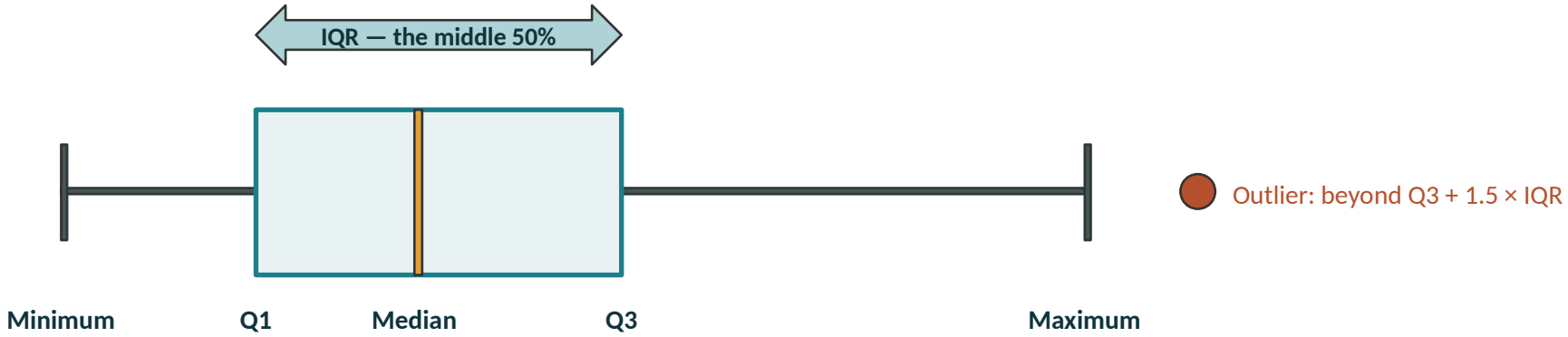
± 3 SD **≈99.7%**

Anything outside is a genuine outlier

z = (x − mean) ÷ standard deviation

Reading a box plot

Five numbers, one picture, instant comparison



What it shows at a glance

Centre, spread, skew and outliers in one figure — no distribution assumption needed.

Where it beats a histogram

Comparing several groups side by side: four box plots fit where one histogram would.

Reading skew

A median sitting off-centre inside the box, or one whisker much longer, means the data are skewed.

Building it in Excel

Select the data, then Insert › Charts › Box and Whisker.

3

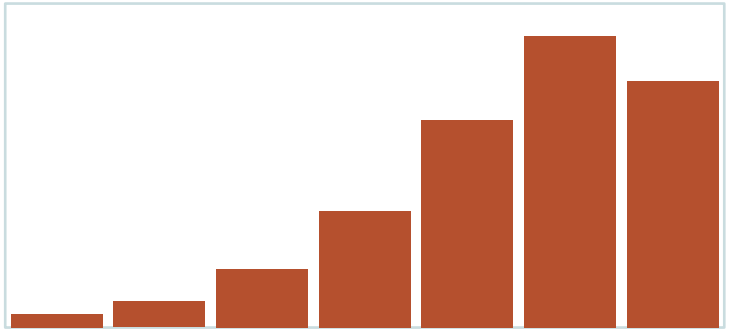
The Shape of a Distribution

Skewness, kurtosis and the observations that do not belong

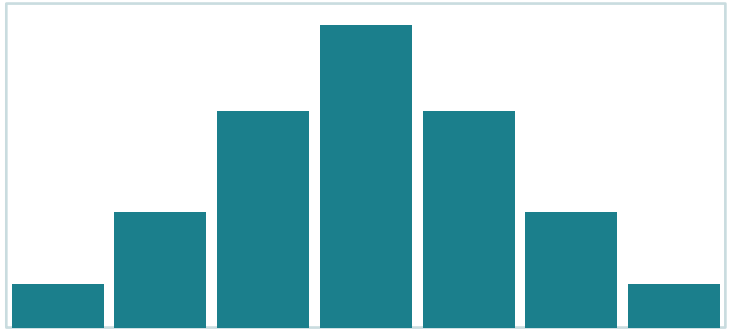
Skewness

Which tail is longer — and what it means for the business

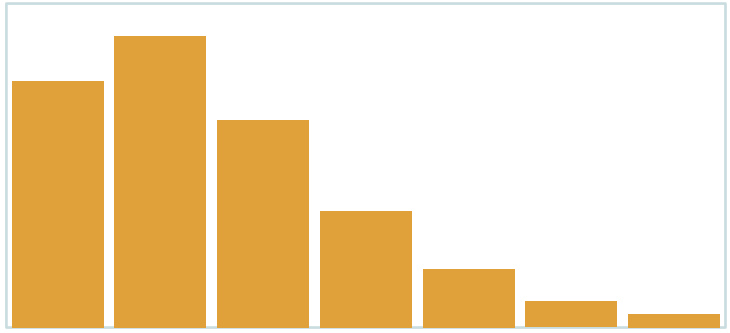
Left-skewed



Symmetric



Right-skewed



Mean < Median

Skewness < 0. A long left tail. Exam scores with a ceiling; delivery times with a hard minimum.

Mean ≈ Median

Skewness ≈ 0. Measurement error, heights, machine-filled weights.

Mean > Median

Skewness > 0. Income, house prices, insurance claims, customer spend — the business default.

=SKEW(range) in Excel. Values beyond ±1 signal substantial skew — report the median alongside the mean.

Kurtosis and outliers

How heavy are the tails, and which points are genuinely unusual?

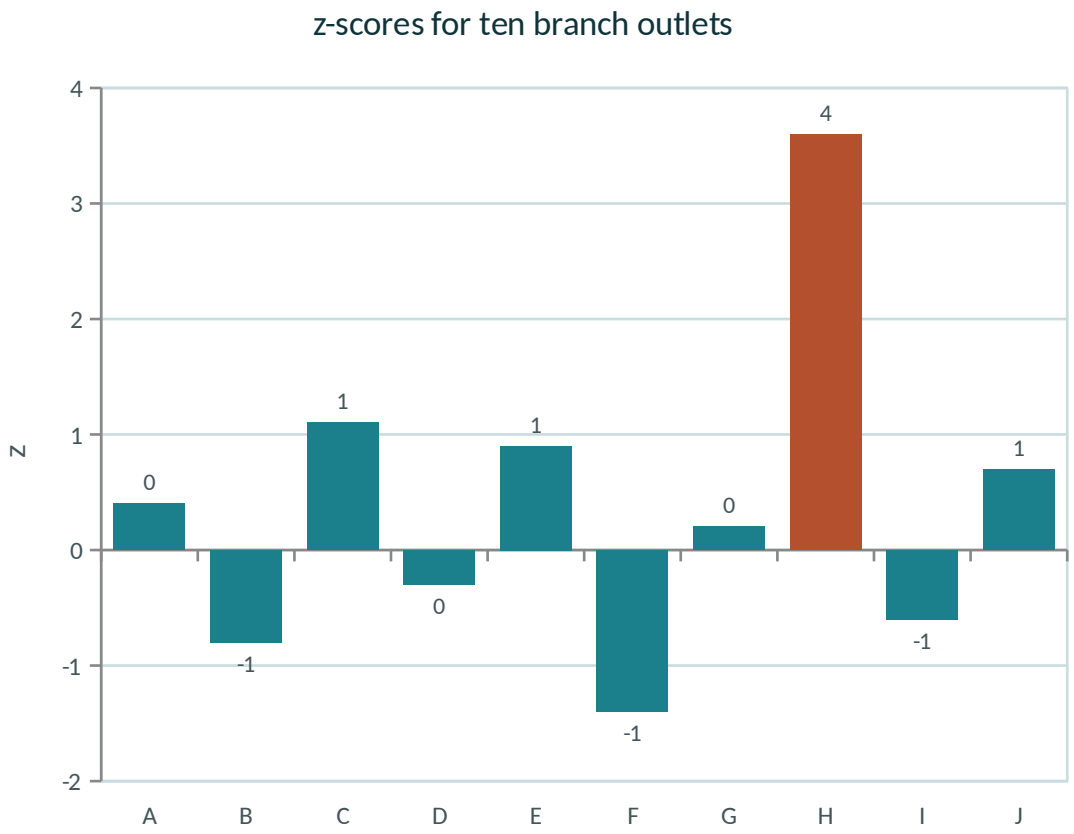
KURTOSIS

Measures how much of the variance comes from rare, extreme observations rather than from typical ones.

Excess kurtosis > 0 → heavy tails: extreme events are more common than a bell curve predicts. Financial returns behave this way, which is why risk models built on the normal curve understate losses.

SPOTTING AN OUTLIER

- z-score rule: |z| > 3 — more than three standard deviations from the mean
- Box plot rule: below Q1 − 1.5×IQR or above Q3 + 1.5×IQR
- Investigate before you delete. An outlier is often the most informative record you have.



Outlet H sits 3.6 SD above the mean — flag and investigate, do not silently drop.

4

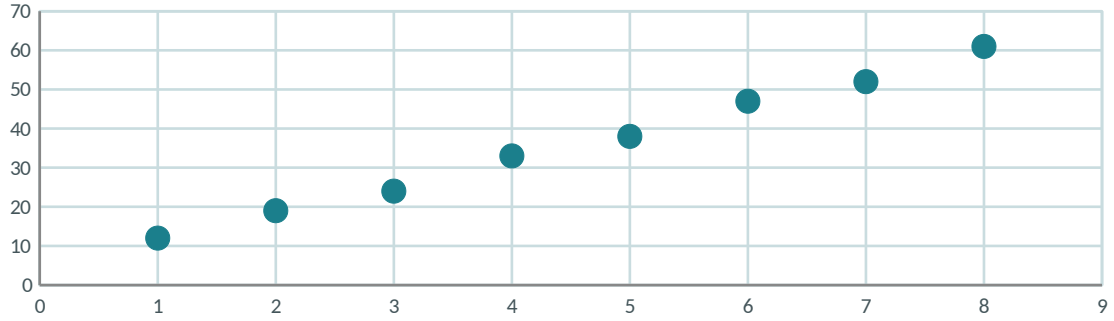
Correlation Analysis

Measuring how two variables move together — and what that does not prove

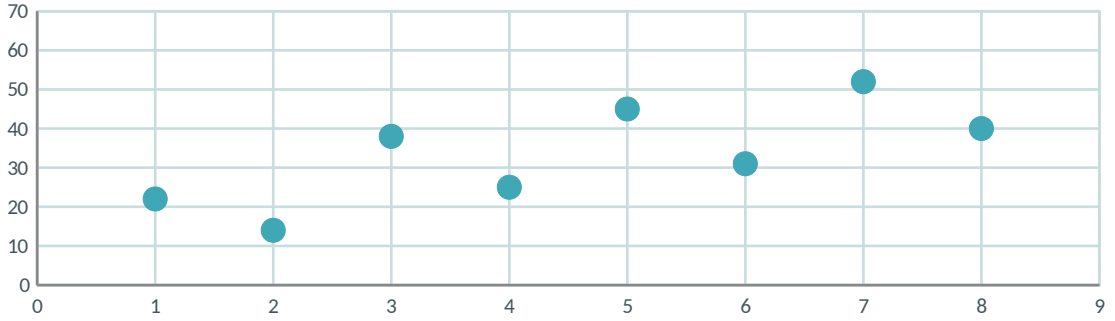
What a scatter diagram shows

Direction, form and strength, before any number is computed

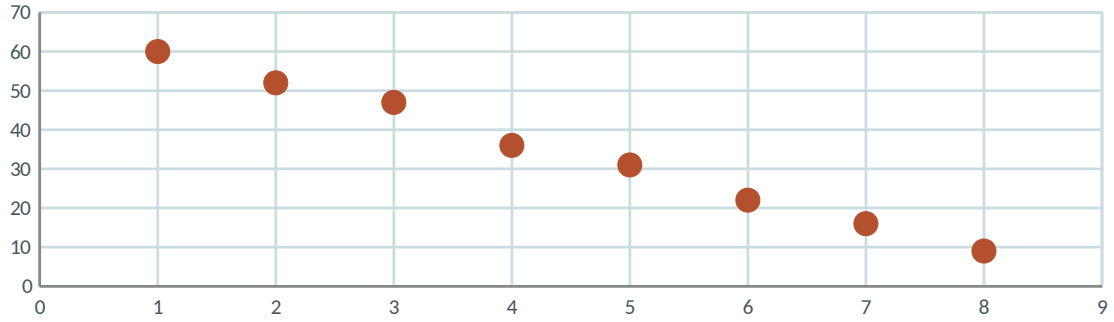
Strong positive r ≈ +0.95



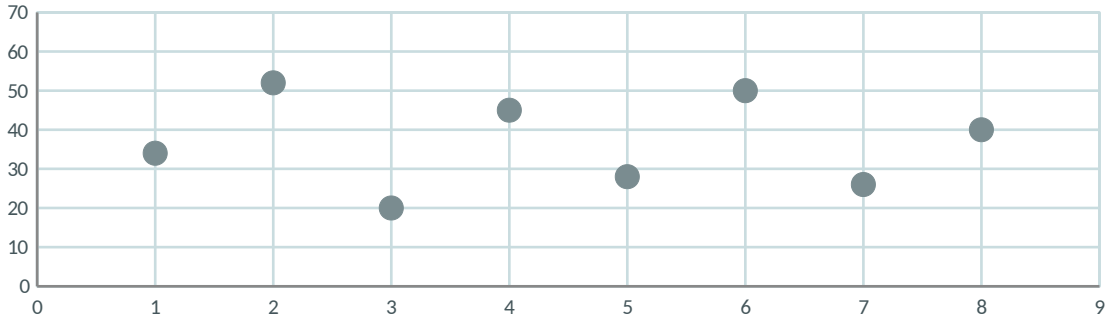
Weak positive r ≈ +0.35



Strong negative r ≈ −0.92



No relationship r ≈ 0



Always plot before you compute. A single scatter reveals curvature, clusters and outliers that r will hide.

Pearson and Spearman

Two correlation coefficients, two different jobs

PEARSON r

Measures the strength of the straight-line relationship between two quantitative variables.

- Always between −1 and +1
- Requires interval or ratio data
- Assumes the relationship is linear
- Sensitive to outliers
- =CORREL(array1, array2)

SPEARMAN ρ

Pearson's formula applied to the ranks rather than the raw values.

- Works with ordinal data — satisfaction ratings, rankings
- Detects any monotonic relationship, not just a straight line
- Resistant to outliers
- =CORREL(RANK array1, RANK array2)

INTERPRETING THE VALUE

−1.0	−0.5	0	+0.5	+1.0
Perfect negative	Moderate negative	None	Moderate positive	Perfect positive

Correlation is not causation

Four explanations for every strong r you will ever see

X causes Y

The relationship you hoped for. Only an experiment can establish it.

Y causes X

Reverse causation. Do busy stores need more staff, or does more staff make stores busy?

A third variable causes both

The classic confounder. Ice-cream sales and drowning deaths correlate — summer causes both.

Pure coincidence

With enough variables, strong correlations appear by chance alone. Test before you believe.

A correlation coefficient describes a pattern in data you already have. Turning it into a decision — ‘if we raise spend by 10%, sales will rise by X’ — requires regression and a defensible research design. That is Module 4.

5

Data Visualisation

One rule: the chart follows the data type, not your preference

Choosing the right chart

Match the question to the picture

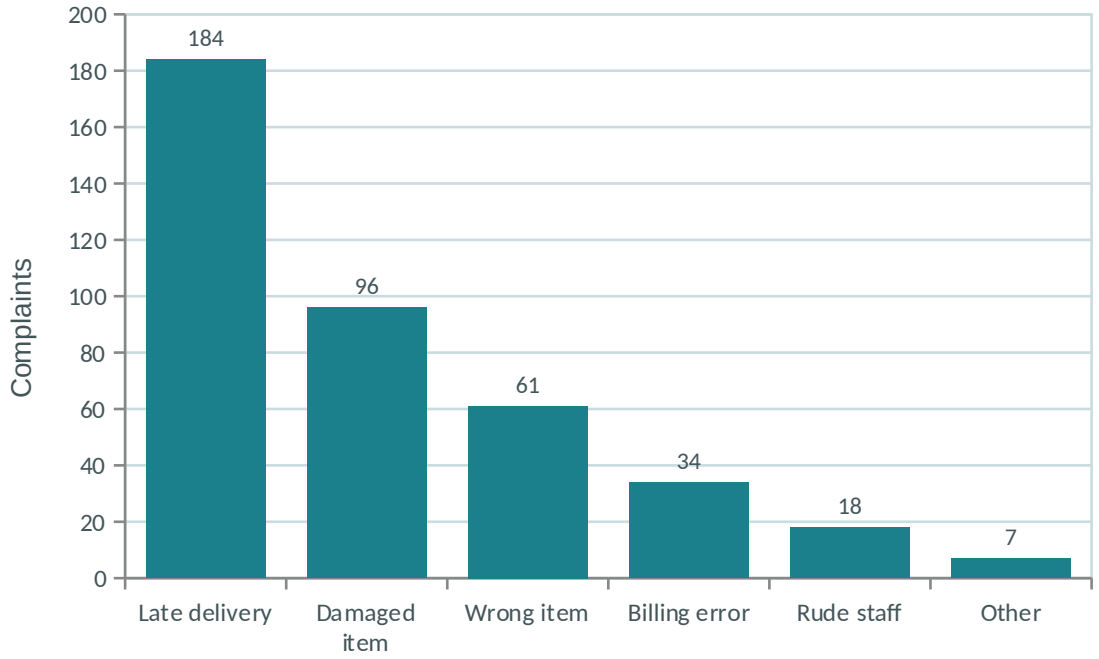
Your question	Data type	Chart	Excel route
How do categories compare?	Categorical	Bar or column chart	Insert › Column
What share does each part hold?	Categorical	Pie chart (≤ 5 slices)	Insert › Pie
Which few causes dominate?	Categorical	Pareto chart	Insert › Histogram › Pareto
How are values distributed?	Quantitative	Histogram	Insert › Histogram
How do several groups compare?	Quantitative	Box and whisker	Insert › Box and Whisker
Do two variables move together?	Two quantitative	Scatter diagram	Insert › Scatter (X,Y)
How has it changed over time?	Time series	Line chart	Insert › Line
How do two categories interact?	Two categorical	Side-by-side / stacked bar	PivotChart

The single most common error in student reports: a pie chart of quantitative data, or a line chart of categories that have no order.

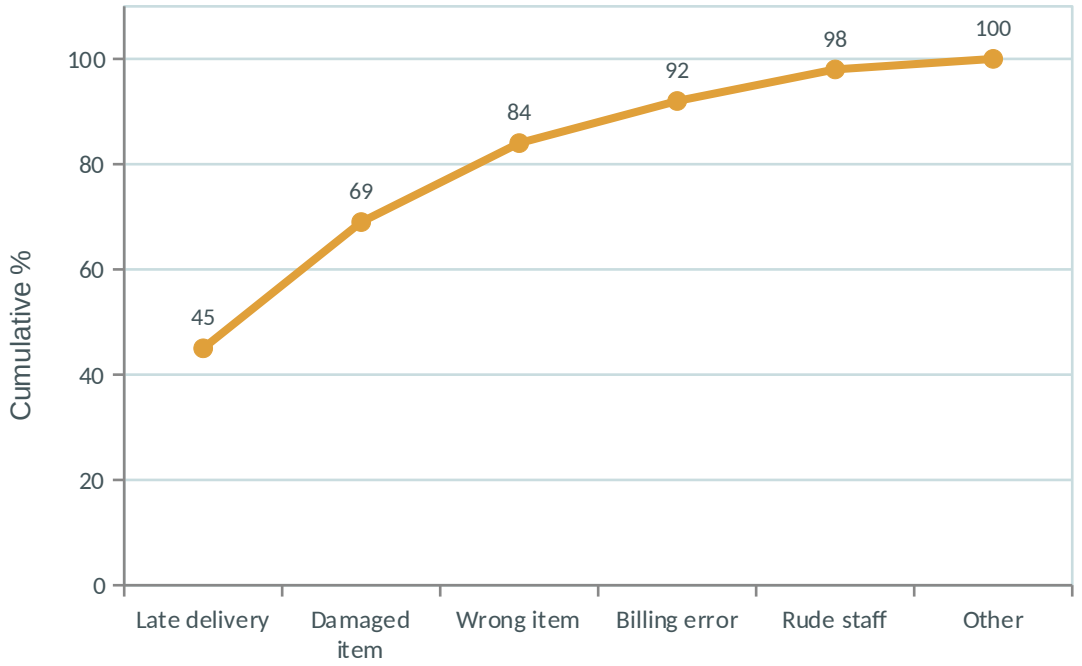
Categorical data: bar, pie and Pareto

The Pareto chart turns a bar chart into a priority list

Customer complaints by cause, sorted descending



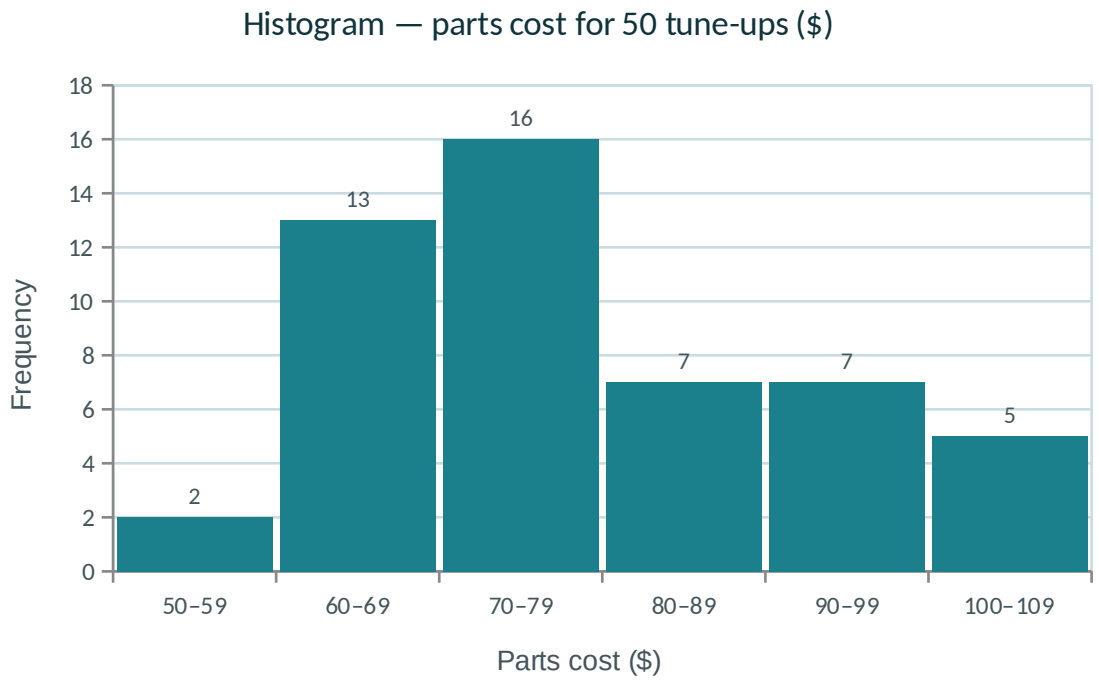
Cumulative percentage of all complaints

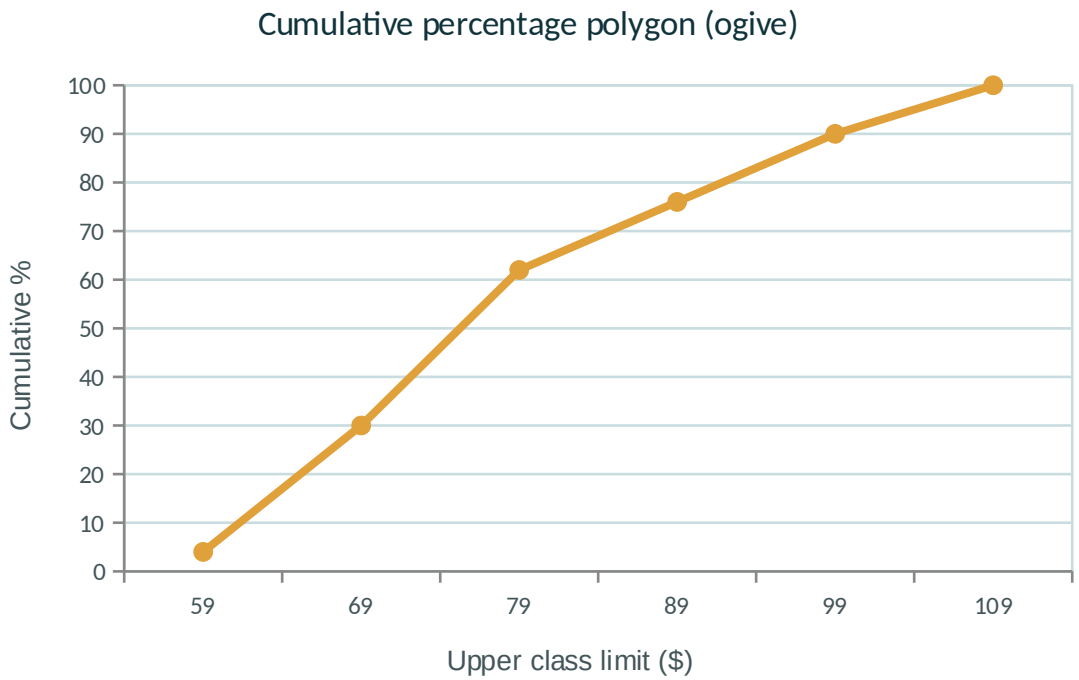


Two causes account for 69% of all complaints. Fixing late delivery and damaged items first is worth more than everything else combined — that is the whole point of a Pareto chart.

Quantitative data: histogram and polygon

Turning a column of numbers into a visible shape





Hudson Auto Repair data, Anderson & Sweeney Ch. 2

Rules of thumb: 5 to 15 classes, all of equal width, with no gaps between the bars. Too few classes hide the shape; too many turn it into noise.

Your Excel toolkit for Module 2

Everything on one page

Task	Function or tool	Where it lives
Mean, median, mode	AVERAGE, MEDIAN, MODE.SNGL	Formulas › More Functions › Statistical
Quartiles and percentiles	QUARTILE.EXC, PERCENTILE.EXC	Formulas › Statistical
Variance and standard deviation	VAR.S, STDEV.S (use .P for a population)	Formulas › Statistical
Shape	SKEW, KURT	Formulas › Statistical
z-score	STANDARDIZE(x, mean, sd)	Formulas › Statistical
Correlation	CORREL, or the Correlation tool for a full matrix	Data › Data Analysis
Everything at once	Descriptive Statistics — tick ‘Summary statistics’	Data › Data Analysis
Grouping and cross-tabs	PivotTable and PivotChart	Insert › PivotTable

If Data Analysis is not visible: File › Options › Add-ins › Manage Excel Add-ins › Go › tick Analysis ToolPak.

Module 2: what to carry forward

Six things you will use in every remaining module

Centre and spread travel together

A mean without a standard deviation is an incomplete answer.

Shape decides the statistic

Skewed data → report the median. Symmetric → the mean is fine.

The SD is the unit of surprise

Anything past ±3 SD is rare enough to investigate.

Plot before you compute

A scatter reveals what a correlation coefficient conceals.

r measures linearity only

A strong curved relationship can still produce r near zero.

The chart follows the data type

Categorical → bar. Quantitative → histogram. Two quantitative → scatter.

Next — Module 3: Inferential Statistics & Hypothesis Testing. Using these same summaries to make claims about a population you never fully observe.

End of Module 2

Practise: compute the full descriptive summary for any dataset of your choice using Data › Data Analysis › Descriptive Statistics, then build a histogram and a box plot from the same column.

Linked assessment: Assignment 1 · Internal Assessment 1